

The origin of *do*-support: a learning-based account

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The goal of the talk

Goal

Explain how *do*-support first emerged, from a **learning-based approach to change**

1. Background: *do* in present-day English
2. Background: the learning-based approach to change
3. Overview of the story
(continued...)

Background: *do* in present-day English

Separating the syntax and semantics

- *Do* has different uses
- I'll organize the uses along two separate axes: syntax and semantics
- Syntactically: auxiliary vs. main verb
- Semantically: varied

- *Do*-support

- (1)
- a. I loved him. (affirmative declarative)
 - b. I **did** not love him. (negative declarative)
 - c. **Did** you love me? (inversion)
 - d. Who **did** you love? (inversion)

- Syntactically, auxiliary
- Semantically, vacuous

- *Do* in VP-ellipsis

(2) John loved Mary, and Peter **did** too.

- Syntactically, ambiguous, but analyzed as auxiliary
- Semantically, vacuous

- Emphatic *do* in affirmative declaratives

(3) I **did** love him.

- Syntactically, auxiliary
- Semantically, emphasis

- Main verb *do*

- (4)
- a. She **did** the dishes.
 - b. She **did** me harm.
 - c. She **did** lots of damage.
 - d. She **did** that.

- Syntactically, main verb
- Semantically, “perform/carry out”, and some others

Background: the learning-based approach to change

The view of learning

- The view of learning:
 1. A child acquires a rule iff they receive sufficient evidence for it
 2. A child builds a grammar by incrementally acquiring rules that build on each other
- A concrete criterion for sufficiency:

(5) The Tolerance Principle (TP; Yang 2016)

Let R be a rule applying to a domain of size N . Let M be the number of items in the domain following R that a learner encounters in their input. Then, the learner generalizes rule R in its domain if and only if $M > N - \frac{N}{\ln N}$.

How change occurs

- Since children acquire a grammar based on their input, their grammar may be different from the grammar(s) that generated their input.

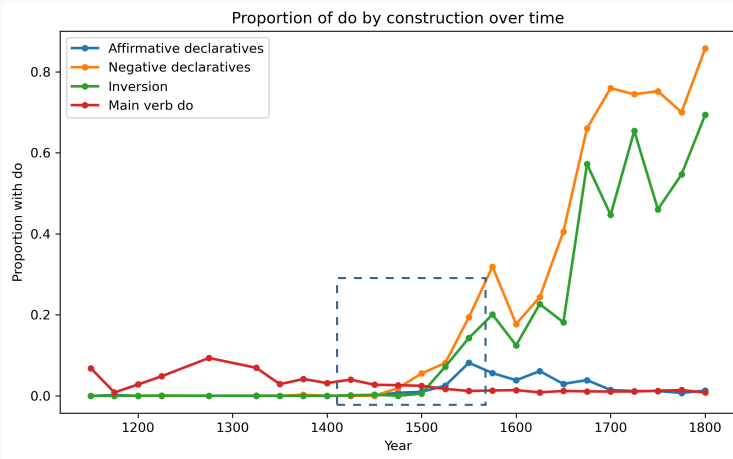
Implications for the historical linguist

- What constitutes an explanation for change?
- Reasoning about what a child can acquire from their input.
 - See Kodner 2020 for an overview

- Penn Parsed Corpora of Historical English (PPCHE; 2025). Includes PPCME2, PPCEME, and PPCMBE2. (Kroch 2020)

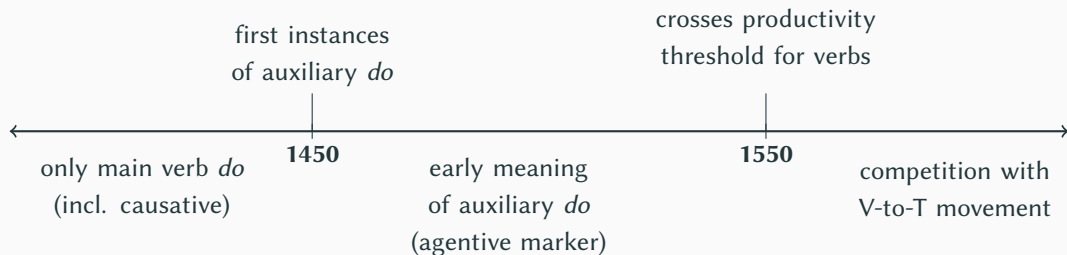
Overview of the story

The history of *do*



- Note: verb movement is lost after *do*-support emerges (Haeberli & Ihsane 2016), thus it doesn't explain the emergence of *do*-support (contra Kroch 1989)

Timeline of early developments



4. Before 1450: the origins of auxiliary *do*
5. 1450–1550: early meaning of auxiliary *do*
6. After 1550: *do*-support

Before 1450: the origins of auxiliary *do*

The puzzle

- Before 1450, all uses of *do* were as a main verb
- Around 1450, we see the first unambiguous instances of *do* as an auxiliary

The puzzle

How did auxiliary *do* emerge from main verb *do*?

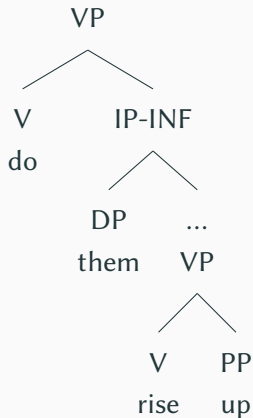
- Or: what evidence does a learner have to reanalyze existing surface strings as underlyingly an auxiliary *do* structure?
- To answer this, let's look at *do* before 1450

Evidence from causative *do*

- *Do* as a causative head

(6) he **dop** hemm risenn upp
he does them rise up
'he causes them to rise up'
(CMORM-M1,I,252.2050)

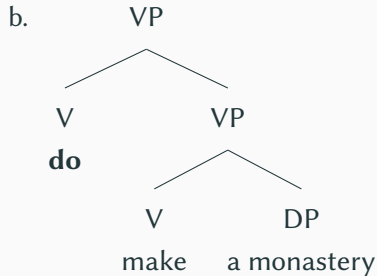
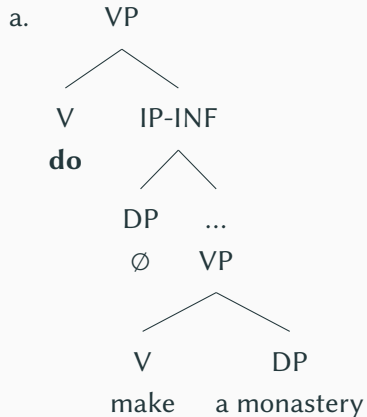
- In my data, these are identified by *do* followed by an IP-INF sister
 - A total of 174, some false positives.
Lost around 1500.



- Ellegård (1953) argues that this is the source of auxiliary *do*
 - Observation: causative *do* can appear without an embedded subject
- (7) And anon he **ded** make a monasteri of þe Cistewis
and at-once he did make a monastery of the Cistercians
'and at once he caused a monastery of the Cistercians to be made'
(CMCAPCHR-M4,91.1799)

Evidence from causative *do*

(8) he **did** make a monastery



Evidence from causative *do*

The problem

Causative *do* alone doesn't provide enough evidence for a child.

- The problem, strictly: For a child to learn that *do* takes a VP, they need to encounter $M > 79$ verbs (in the top 100) occurring with *do*¹
 - Our data has 25 verbs of the 100, no coherent semantic class
 - Unfortunately, we won't be able to solve this problem here
- The problem, loosely: The child has no reason to analyze subjectless causative *do* as anything but that
 - We will make some progress on this problem
 - To do so, we need to find new sources of evidence

¹For $N = 100$, $N - \frac{N}{\ln N} = 79$.

Evidence from anaphoric *do*

- Another source: anaphoric *do* is often compatible with *do* as a main verb or auxiliary

(9) a. and he shal gete as miche as his auncestres **deden**
and he shall get as much as his ancestors did
'and he shall get as much as his ancestors did'
(CMBRUT3-M3,75.2261)

(In

b. and freure us of alle sorege also he hem **dide**
and comfort us of all sorrow as he them did
'and comfort us from all sorrow, as he did them'
(CMTRINIT-MX1,121.1617)

PDE we analyze these as auxiliary *do* + ellipsis, but these examples precede clear auxiliary *do*)

- Not enough evidence on their own, but another source of syntactic ambiguity, in addition to causatives

Evidence from performance *do*

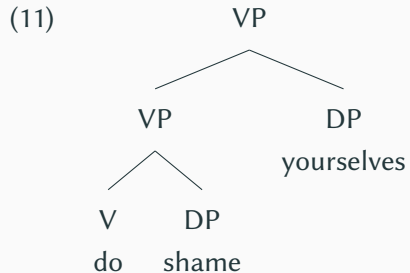
- Lastly, consider “performance” *do* (takes DP-complement, means “perform/carry out”)

(10) ʒie ... **doð** scame ʒew seluen
you do shame you selves
‘You bring shame upon yourselves.’
(CMVICES1-M1,123.1492)

- Semantically, *do* takes an event-denoting object, and introduces the agent
- Syntactically, compatible with two structures

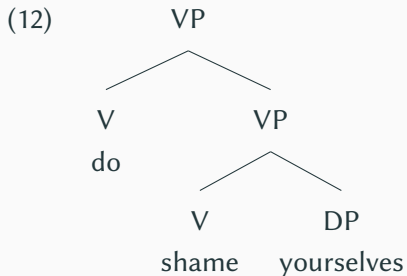
Evidence from performance *do*

- The conservative structure: main verb *do* + DP-complement



Evidence from performance *do*

- The innovative structure: auxiliary *do* + VP-complement



- This is made possible by:
 1. Loss of *-en* infinitive
 2. Loss of case (+ merging of dative and accusative in pronouns)
 3. Dative objects can be bare DPs, not yet obligatorily PPs

- These examples are too rare to be the primary source of syntactic reanalysis, but can be part of the story...

Putting the pieces together

- Semantically, children learn performance *do*: *do* introduces the agent to an event denoted by its object.
- Syntactically, in the conservative grammar, this object is a DP. Thus *do* is a main verb.
- However, due to aforementioned factors, this complement is ambiguous between a DP and VP.
- This, reinforced by evidence from causative and anaphoric *do*, motivates a generalization that *do* can take VPs. This is auxiliary *do*.

**1450–1550: early meaning of
auxiliary *do***

Do as an agentive marker

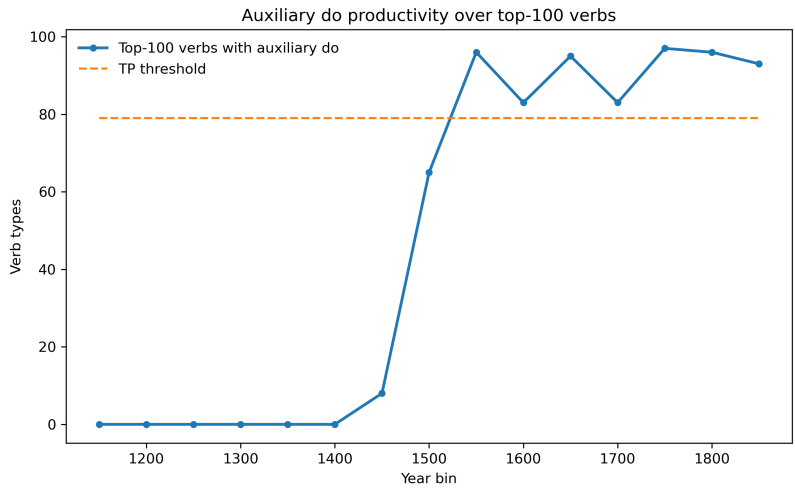
- Ecay 2015: early auxiliary *do* as an agentive marker
 - Supported by the rarity of *do* with unaccusatives
 - Consistent with Ellegård's observation: auxiliary *do* is initially resisted by a class of verbs (e.g. *know*, *doubt*, *fear*)
- What is this “agentive marker”?
 - Syntactically, auxiliary *do*
 - Semantically, *do* introduces the agent of the event
- Continuity with the result from the previous section

Conditional productivity of agentive *do*

- By nature, agentive *do* cannot be used with all verbs
- I.e., it is only productive within a semantic class (e.g. those with agentive subjects)

Unconditional productivity of *do*

- Somehow, auxiliary *do* becomes unconditionally productive by 1550

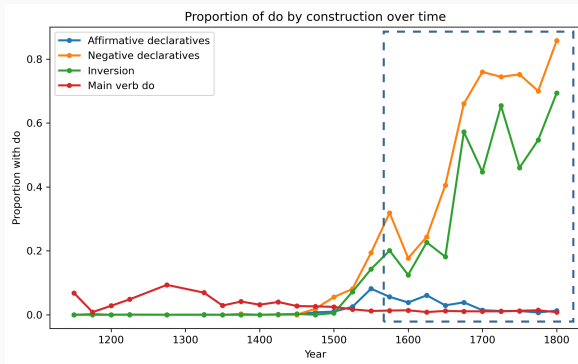


- Once auxiliary *do* starts appearing with unaccusatives/statives, its meaning as an agentivity marker is no longer learnable
- In fact, no semantic generalization may be learnable (“semantic bleaching”)
- This gives us *do*-support: syntactically an auxiliary, but semantically vacuous

After 1550: *do*-support

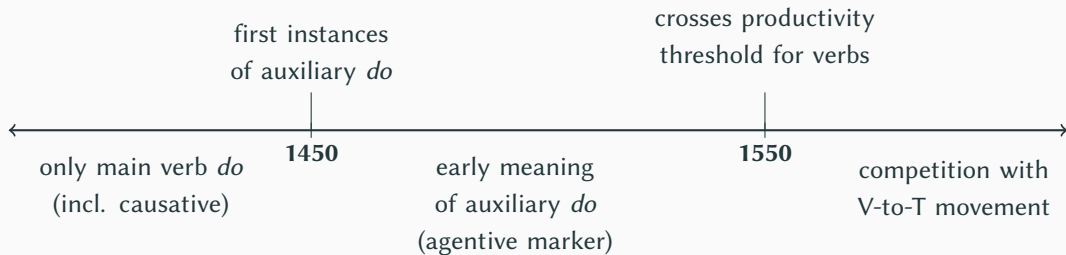
Entering into the competition

- Now, sentences with *do*-support have the same meaning as those without
- Thus *do*-support starts to directly compete with the existing structures
 - Loses in aff. decl. (or narrows in meaning)
 - Wins in other environments
 - More work needed (see Heycock & Wallenberg 2013 for Scandinavian)



Conclusion

Summary



- In this talk, I have
 - proposed a learning-based story for the origin of auxiliary *do*, combining evidence from various uses of *do*
 - shown how this can lead to Ecay's “intermediate” *do*
 - shown that *do* becomes unconditionally productive around 1550
 - connected this to semantic bleaching and competition with V-to-T movement

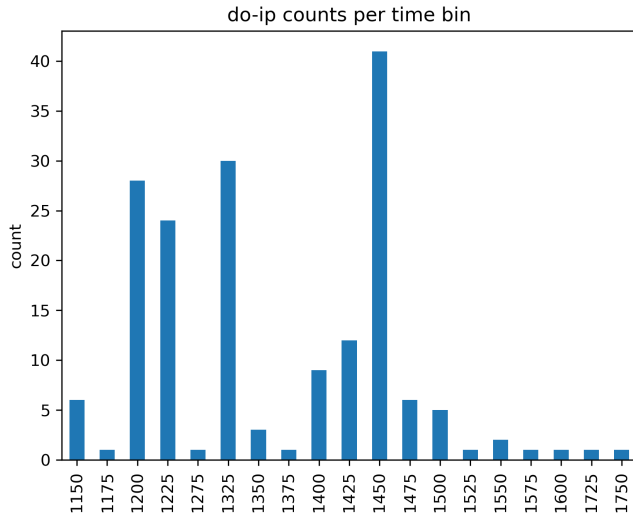
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- I thank Charles Yang, Don Ringe, and Beatrice Santorini for helpful comments
- Slides are available on my website: liankwang.github.io

Appendix

Distribution of causative *do*



Case, infinitives, datives, word order

- Largely as a consequence of phonological weakening of final unstressed syllables, loss of many endings (e.g. infinitive *-en*, case endings, etc.)
- No acc/dat for regular nouns, acc/dat merged for pronouns
- Infinitive suffix *-en* was lost, thus we get new verb forms like *harm*, *wronge*, *sin*, etc.
- Once dative case is lost, “dative” objects are formally identical to accusatives. But they are still productively adjoined to VPs (unlike in PDE, where additional arguments must be introduced by P), and can appear in either order

Case, infinitives, datives, word order

- (13) Dative introduced by preposition
and **dede** miche sorwe to *þe Britons*
and did much sorrow to the Britons
'and caused much sorrow to the Britons'
(CMBRUT3-M3,38.1188)
- (14) Dative as bare noun
and freure us of alle sorege also he *hem*
and comfort us of all sorrow as he them
dide
did
'and comfort us from all sorrow, as he did them'
(CMTRINIT-MX1,121.1617)
- (15) Dative following direct object
doð scame *zew seluen*
do shame you selves
'do shame to yourselves'
(CMVICES1-M1,123.1492)
- (16) Dative preceding direct object
Drihtin **do** *me* merci
Lord do me mercy
'Lord, show me mercy'
(CMMARGA-M1,92.588)

Case, infinitives, datives, word order

- Objects of transitives can appear before the verb

(17) a. al duden þat cheorles swa Cador *heom* **tæhte**
all did the churls as Cador them taught
'The churls all did as Cador taught them'

(1200-BRUT-M1,560.1263)

b. þatt mann *himm* **band** wiþ woðhe
that man him bound with wrong
'that man bound him unjustly'

(CMORM-M1,I,39.415)

- This may give new sources of structural ambiguity in causatives, if the meaning works out

Examples of early auxiliary *do* in 1450

- (18) a. he **ded** crye openly
he did cry openly
'he proclaimed openly'
(CMCAPCHR-M4,40.257)
- b. but ye **do** overmacche hym
unless you do overmatch him
'unless you overmatch him'
(CMMALORY-M4,189.2744)
- c. all the trouth sir Gahalantyne **dud** telle
all the truth Sir Gahalantyne did tell
'Sir Gahalantyne told all the truth'
(CMMALORY-M4,209.3472)

